CSA4707-Deep Learning

Experiment 5: Linear Regression

Aim:

To demonstrate the performance of linear regression using a chosen dataset with Python.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

iris = load_iris()

data = pd.DataFrame(
    data=iris.data, columns=iris.feature_names
)
data['species'] = iris.target

print(data.head())

X = data[['sepal length (cm)']]
y = data['sepal width (cm)']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
print(f'Mean Squared Error on test set: {mse:.2f}')

plt.figure(figsize=(10, 6))
plt.scatter(X_test, y_test, color='r', marker='*', label='Actual')
plt.plot(X_test, y_pred, color='y', linewidth=3, label='Predicted')
```

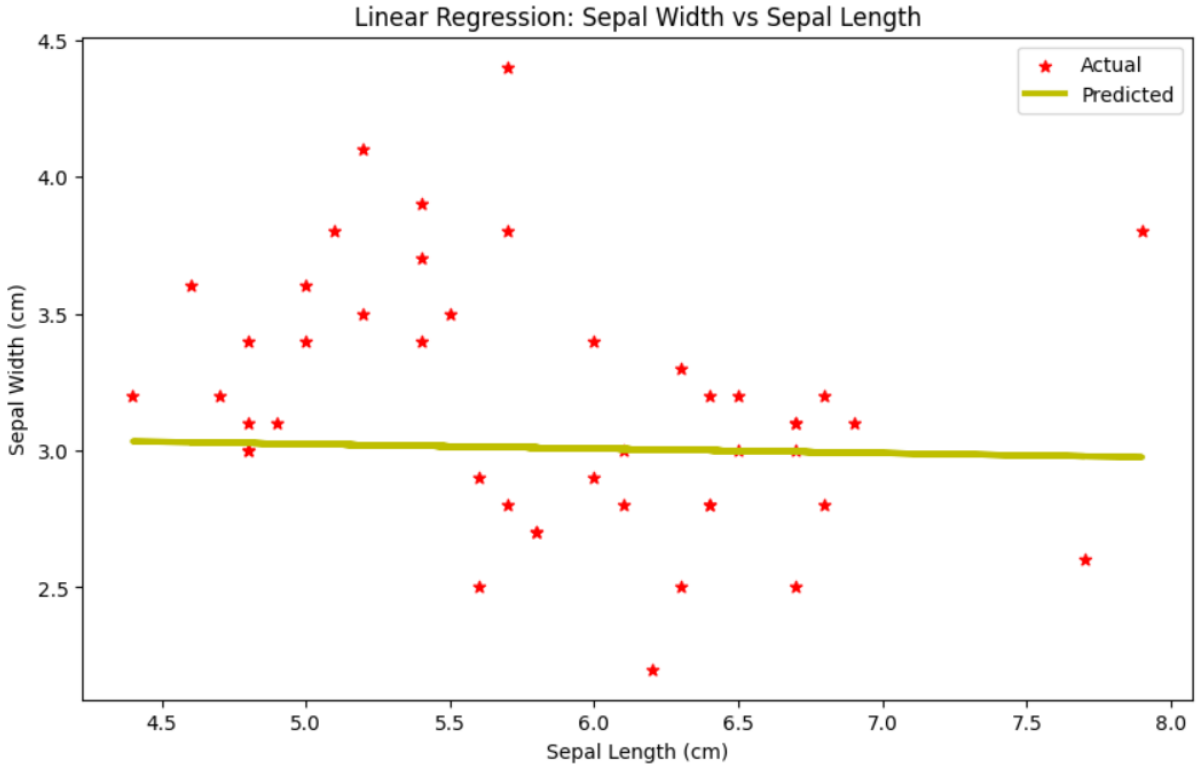
```
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Linear Regression: Sepal Width vs Sepal Length')
plt.legend()
plt.show()

new_sample = pd.DataFrame(
    [[5]], columns=['sepal length (cm)']
)

predicted_width = model.predict(new_sample)

print(
    f'The predicted sepal width for sepal length '
    f'{new_sample.values.tolist()} is {predicted_width[0]:.2f} cm'
)
```

Output:



Result / Output:

Mean Squared Error on test set: 0.23

The predicted sepal width for sepal length [[5]] is 3.02 cm

The linear regression model was trained successfully to predict sepal width from sepal length. The predicted sepal width for a sepal length of 5 cm is approximately 3.02 cm.